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Spatiotemporal analysis of mangroves using median composites and convolutional neural network

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Mangroves play an important ecological role, but these are commonly misunderstood and undervalued. Climatic change and increase in sea level cause risk to these ecosystems. Hence, tracking the extent of mangroves leads to proper management and restoring the damaged ones. Existing index methods for mapping the mangrove extent often struggle with cloud cover, while the studies that feed direct image inputs to models face spatial resolution misalignment, limiting accurate analysis. To overcome these challenges this paper proposes a novel cloud-masked feature extraction (CMFE) approach, integrating bit masking, median compositing, and multi-band spectral sampling for precise mangrove analysis. Our approach utilizes multispectral imagery from United States Geological Survey (USGS) for Landsat 8 and Copernicus for Sentinel 2, leveraging quality assurance (QA) bands for cloud bit masking to nullify cloud and shadow pixels. Cloud cover obscures large portions of satellite imagery and reduces data availability. The median composite technique reconstructs missing or obscured information by aggregating temporal pixel statistics. Spectral and textural attributes are systematically extracted from multiple bands and structured into feature vectors, providing high-dimensional input, which overcomes the spatial resolution discrepancies. Extracted attributes are then fed to a deep learning-based model which classifies the mangroves into different density classes - sparse, moderate, and dense for mapping the dynamic changes over a decade from 2014 to 2024, focusing on Pichavaram Mangrove Forest, Tamil Nadu. The CMFE approach outperforms existing techniques, offering improved accuracy and resilience against cloud cover and spatial misalignment.

Keywords Remote sensing, Satellite imagery, Spatiotemporal analysis, Mangrove mapping, Bit masking, Convolutional neural network

Mangroves play a vital role in coastal ecosystems, serving as an interface between terrestrial and marine environments while providing natural protection against floods, waves, and strong winds[1,2]. These habitats also enhance water quality by filtering pollutants, trapping sediments, and reducing coastal erosion[3,4]. However, mangrove ecosystems are highly dynamic and susceptible to both natural and human-induced changes. Climate change, tidal variations, and natural disasters contribute to fluctuations in mangrove cover, while deforestation, pollution, and coastal development accelerate their degradation[5]. Over the past century, approximately 75% of the world's mangrove forests have been lost, with studies estimating a 3.4% global decline from 1996 to 2020[6]. Therefore, accurate spatial and temporal mapping, along with continuous monitoring of mangrove ecosystems, is crucial for conservation efforts and sustainable development[7,8].

Remote sensing technology is widely used for mangrove monitoring as it provides fast information acquisition and, an efficient and cost-effective approach[1,9]. With the help of remote sensing technologies, conservation strategies[10] can be developed easily as these offer valuable information about the condition and evolution of mangrove forests[11]. Mainly multispectral images, hyperspectral images, Light Detection and Ranging (LIDAR), and Synthetic Aperture Radar (SAR) datasets have been used for mangrove studies[8], with long-term Landsat imagery on Google Earth Engine revealing significant land use changes around mangroves in Thailand's Trat Province[12]. Even though SAR images are not affected by cloud cover[13], optical images offer high resolution for detecting subtle changes in mangrove density, which is crucial for precise classification. Some studies use hyperspectral data for mapping mangroves but the major problem with hyperspectral is that it is not used for most mangrove areas and cost prohibitive for monitoring large scale[14].

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The classification of mangroves can be done in two ways – index-based thresholding method and machine learning models[15]. The Remote Sensing for Mangrove Identification (REMI) method leverages specific satellite bands, such as red, red edge, green, and SWIR1 to effectively capture distinctive mangrove characteristics[1]. The Generalized Composite Mangrove Index (GCMI) uses the vegetation index and water index where the water index is used to highlight the moisture content of vegetation and the vegetation index is used to distinguish the mangroves from other vegetation. Combining these indices improves the separability between mangroves and other land covers[7]. These index-based methods often misclassify non-mangrove trees as mangroves and exhibit instability in non vegetation areas hindering the accuracy and complexity to learn complex spatial characteristics and focus on the spectral information[16,17]. Cloud masking techniques are performed to remove the cloud along with a deep learning method for generating the Landsat 5 images with 10 m resolution[18].

Machine Learning algorithms such as Random Forest, Support Vector Machine, and Deep Learning play a major role in the mapping of mangroves[19–21]. Object-based Image Analysis (OBIA) and Random Forest were integrated to map mangroves achieving high accuracy despite challenges in mapping small mangrove fragments and dataset-specific detection issues[3]. The integration of the various datasets, including Gaofen-2, Gaofen-3, and UAV-LiDAR data for comprehensive geospatial analysis and applied the Random Forest Regression to retrieve height information is demonstrated in[5]. Satellite images, mangrove ecosystems in the Hara protected area, Qeshm, Iran, were accurately mapped with high precision through the integration of seasonal optical and Synthetic Aperture Radar (SAR) features[8]. Sentinel 1 and Sentinel 2 red edge band data were used to evaluate the mangrove mapping efficiency using Random Forest by comparing three scenarios: Sentinel 1, Sentinel 2, and the combined data. A limitation observed in this method is ineffectiveness of images in identifying and monitoring isolated mangrove forest patches smaller than 1 Hectare (ha)[10]. Using Sentinel-1 and Sentinel-2, the Google Earth Engine Mangrove Mapping Methodology (GEEMMM) uses tidal calibration and cloud computing to enable global mangrove mapping. However, scaling to bigger areas proved to be difficult[19]. Another technique involves using Sentinel 1 for multitemporal composite analysis and Sentinel 2 for land cover classification with Support Vector Machines for glacier mapping[22]. Random Forest and Support Vector Machine algorithms highlighted suboptimal performance due to the neglect of spatial dependencies among adjacent pixels[2], while GEE-based tools like MANGLEE mapped mangrove losses in Ecuador due to shrimp farming[23] and fused pixel- and object-based classification improved LULC mapping with Landsat 8[24].

Random Forest (RF) and Support Vector Machines (SVM), are widely used Machine learning (ML) algorithms for mangrove mapping. But, these models have limitation in accurately capturing the relationships between spectral and spatial attributes that are critical for distinguishing mangrove densities. These techniques struggle with high-dimensional data and having difficulty in identifying small mangrove patches, leading to suboptimal performance. ML models tend to neglect spatial dependencies between adjacent pixels, which reduces their effectiveness in mapping mangroves in highly dynamic environments affected by factors such as cloud cover.

To overcome these limitations our study leverages a deep learning-based approach, specifically using a Convolutional Neural Network (CNN) integrated with Cloud-Masked Feature Extraction (CMFE) as represented in the Fig. 1. This method extracts both spectral and textural attributes from multiple bands, creating high dimensional feature vectors that capture the nuances of mangrove density. By feeding these features into the CNN, our model effectively classify mangroves into sparse, moderate, and dense categories. The median compositing technique reconstructs missing data in cloud-covered regions, and the deep learning model learns complex patterns that ML models struggle to capture, resulting in superior accuracy in mapping mangrove density over time. This makes our approach far more resilient and precise than traditional ML methods.

Study area and dataset

Study area

Pichavaram Mangroves are the second largest mangrove forest in India. These are one of the most productive ecosystems and act as a storehouse of biodiversity. It is located in Cuddalore district, Tamil Nadu state, and covers up to 45 km^2 of the area. These mangroves act as a crucial buffer zone, mitigating coastal erosion and storm surges while supporting a rich biodiversity of floral and faunal species. The region experiences a tropical climate,

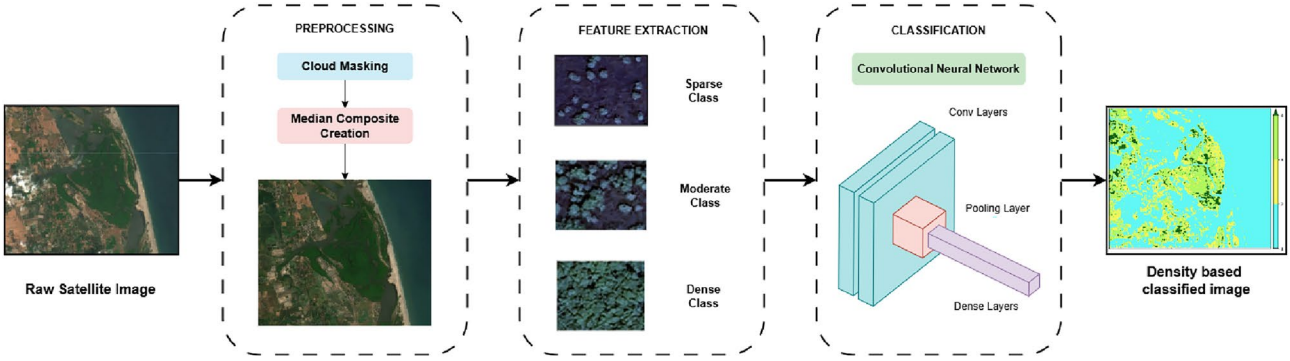


Fig. 1. Workflow Diagram for Health Assessment. Map generated using Google Earth Engine (Web-based, accessed 2023–2024, https://earthengine.google.com/).

with an annual rainfall of 1200 mm, primarily during the monsoon season, and is influenced by semi-diurnal tides that regulate water salinity and nutrient flow. The mangrove density in this region varies significantly, with sparse patches found along tidal inlets and dense stands observed in the protected core areas.

The Pichavaram wetland is subject to both natural and anthropogenic pressures, including aquaculture expansion, tourism, and pollution, which threaten mangrove sustainability. Conservation efforts by local authorities and research institutions focus on restoration and sustainable management.

Multispectral satellite imagery

Multispectral satellite imagery provides useful data across a spectrum of wavelengths that includes visible, near-infrared, and shortwave infrared bands. Each band captures distinct information on surface properties and precise differences among vegetation, water, soil, and built environments. This capability allows detailed environmental analysis and change detection. In this study, we utilize data from Landsat 8's Operational Land Imager (OLI) and Sentinel-2's Multispectral Instrument (MSI), both obtained from Google Earth Engine (GEE).

For Landsat 8 OLI Level-2 Surface Reflectance (SR) products were initiated by the United States Geological Survey (USGS), which are processed using the Land Surface Reflectance Code (LaSRC). This preprocessing includes atmospheric correction, radiometric calibration to ensure accurate surface reflectance values. Landsat 8 provides 11 spectral bands, with a spatial resolution of 30 meters. These images contain 5 visible and near-infrared (VNIR) bands and 2 short-wave infrared (SWIR) bands processed to orthorectified surface reflectance, and one thermal infrared (TIR) band processed to orthorectified surface temperature.

Sentinel-2 MSI Level-2A products are part of the Copernicus program of the European Space Agency, which was processed using the Sen2Cor algorithm. This includes atmospheric correction and terrain correction, and offers 13 spectral bands, with pixel sizes of 10 m (visible and NIR), 20 m (red-edge and SWIR), and 60 m (aerosol and water vapor bands), allowing for detailed mangrove mapping and change detection.

Data collection

The study area for the Pichavaram Mangroves is defined by a set of georeferenced coordinates as: $11^\circ 38' 19''$ N - $11^\circ 38' 05''$ N, $11^\circ 12' 04''$ N - $11^\circ 12' 14''$ N latitude and $79^\circ 56' 16''$ E - $79^\circ 38' 28''$ E, $79^\circ 38' 23''$ E - $79^\circ 56' 31''$ E longitude respectively.

A series of images were acquired between the years 2014 and 2024 given in to capture spatial and temporal changes as mentioned in the Table 1. Our paper utilizes image data from satellites such as Landsat 8's OLI and Sentinel 2's MSI Level 2A. The datasets were curated using GEE to provide a comprehensive view of mangrove dynamics in Pichavaram over the last decade.

Methodology

Data preprocessing-median composite

Performing data preprocessing on raw satellite images with unclear pixels as shown in Figs. 3(a) and 3(c) is crucial for ensuring the quality of collected images. This helps to remove the interference caused by cloud cover and shadows and enhances the clarity of the images. Hence, it is suitable for the accurate analysis of mangrove density.

Data preprocessing performed in our study involves two steps: (1) cloud masking, which eliminates cloud and cloud shadow pixels to reduce atmospheric interference, and (2) median composite creation, which reconstructs image data by considering all median values across multiple images taken within similar time frame.

Cloud removal is a critical step in remote sensing, and various techniques offer distinct advantages and limitations. Interpolation-based cloud removal[25] estimates missing pixel values using spatial or temporal neighboring data. However, this method can introduce artificial variations and may not always maintain spectral integrity. Deep learning-based cloud removal approaches, such as Generative Adversarial Networks (GANs) in[26], predict cloud-free pixels by learning patterns from extensive datasets. While these models can produce better cloud pixel reconstructions, they require significant computational resources and large amounts of training data, and they may introduce spectral inconsistencies. In contrast, bitmasking and median compositing techniques ensure the retention of high-quality, observed data. Cloud masking effectively eliminates cloud-contaminated pixels using predefined spectral bitmasks, while median compositing reconstructs cloud-free images by selecting the median value from multiple time-series observations. This approach minimizes the impact of outliers, preserves spectral fidelity, and provides a computationally efficient, robust, and reliable solution for cloud and cloud shadow removal.

Cloud masking

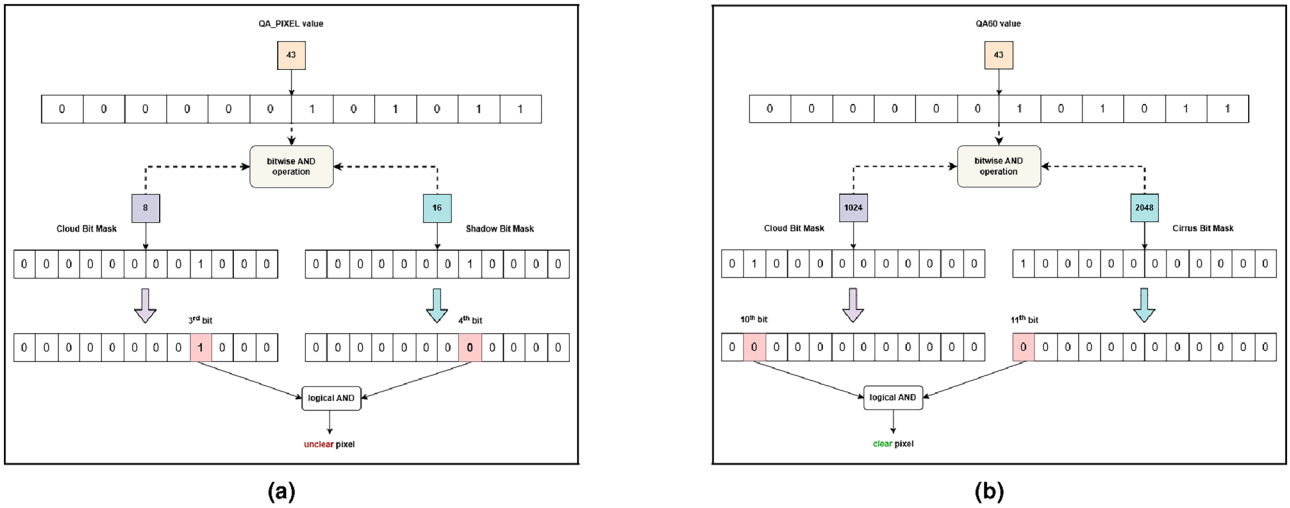
In this step, the special spectral bands as specified in Table 2 are utilized to mask out the cloud and cloud shadow pixels present within the raw satellite image. In QA_PIXEL band, the cloud and cloud shadow pixels flagged in 3rd bit and 4th bit respectively. Whereas in the QA60 band, the cloud and cirrus cloud pixels are flagged in 10th and 11th bit respectively[18].

Time period	Satellite
2014–2016	Landsat 8 OLI
2017–2024	Sentinel 2 MSI level 2A

Table 1. Time period and Satellite image.

Satellite	QA band
Landsat 8 OLI	QA_PIXEL
Sentinel 2 MSI Level 2A	QA60

Table 2. Satellites and their QA bands. Values obtained using the proposed method are highlighted.



(a) (b)

Fig. 2. Bit masking for satellite images (**a**) Landsat 8, (**b**) Sentinel 2.

An image's pixel and its corresponding QA band according to its respective satellite are taken. In this study, bitmasks for cloud presence as cloudBitMask and for cloud shadow as shadowBitMask are defined for Landsat 8 as given in Eqs. (1) and (2).

$$\text{cloudBitMask} = 1 \ll 3 = 8 \tag{1}$$

$$\text{shadowBitMask} = 1 \ll 4 = 16 \tag{2}$$

These values serve as masks for 3rd and 4th bit indicating cloud and shadow conditions. For Sentinel 2, bitmasks for opaque cloud presence as *opaqueBitMask* and for cirrus clouds as *cirrusBitMask* are defined as given in Eqs. (3) and (4).

$$\text{opaqueBitMask} = 1 \ll 10 = 1024 \tag{3}$$

$$\text{cirrusBitMask} = 1 \ll 11 = 2048 \tag{4}$$

These values serve as masks for 10th and 11th bit indicating opaque and cirrus cloud conditions. A bitwise AND operation as shown in Figs. 2(a) and 2(b) is applied between the extracted QA band value and the defined bitmasks according to their satellite. These operations are expressed as given in Eq. (5). If the result of this operation is 0, it indicates a clear pixel, while a result of 1 indicates an unclear pixel.

$$\text{clearFlag} = \text{QA Band} \,\&\, \text{BitMask} \tag{5}$$

Median composite creation

The Median Composite is calculated[18] by sorting pixel values from a set of images taken within a specific seasonal window and selecting the median value for each pixel position shown in Eq. (6).

$$\text{MedianComposite} = \text{sort}\left([i_1, i_2, \ldots, i_n]\right)\left(\frac{n+1}{2}\right) \tag{6}$$

where, i is an image and n is the number of images. After data preprocessing, the composite image obtained is shown in Figs. 3(b) and 3(d).

This median compositing mitigates spatial misalignments by minimizing the influence of temporally inconsistent pixel values. By selecting the median pixel value across multiple images, it suppresses outliers caused by slight misregistrations, resulting in a more stable and representative composite. Thus enhances the reliability of feature extraction and classification by reducing distortions associated with spatial shifts.

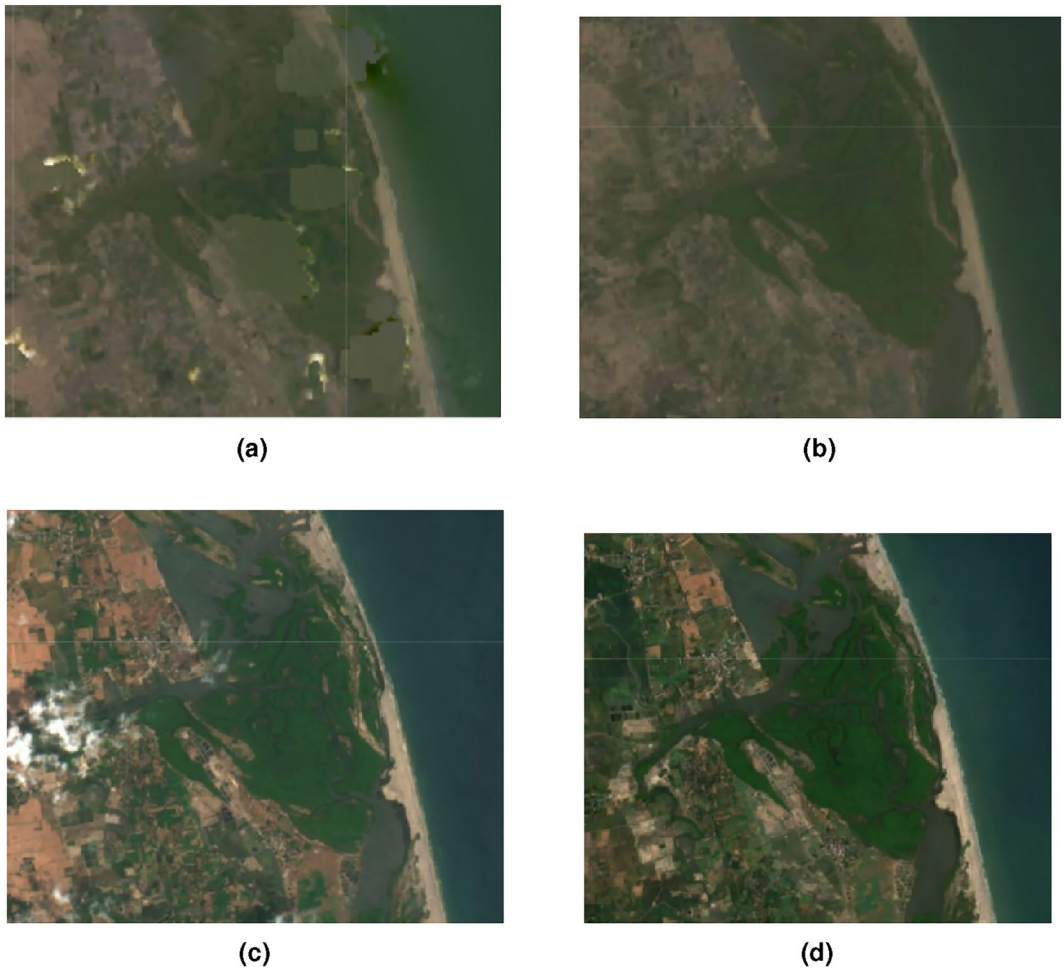


Fig. 3. Before median composite (**a**) Landsat 8, (**c**) Sentinel 2—after median composite, (**b**) Landsat 8, (**d**) Sentinel 2. Maps generated using Google Earth Engine (Web-based, accessed 2023–2024, https://earthengine.google.com/).

Density-based classification of mangroves

Following the creation of median composites, the reflectance values from spectral bands are extracted to accurately classify the mangrove ecosystem. The selected spectral bands, as part of the curated dataset are B1 to B7 for Landsat 8 and B1 to B4, B8, B11, B12 under Sentinel 2 which are necessary for distinguishing density levels within mangroves and accurately classifying mangrove and non-mangrove areas. These bands, chosen for their distinct spectral characteristics, enhance the classification accuracy of mangroves. The B1 (coastal aerosol) is useful for coastal studies. The B2 (blue) band monitor water bodies whereas B3 (green) band differentiates vegetation types. The B4 (red) band captures reflectance that is sensitive to chlorophyll and is useful for observing changes in vegetation health. Landsat 8 uses B5 whereas Sentinel 2 uses B8 for near-infrared, which is essential for assessing vegetation density. The shortwave infrared (SWIR 1 and SWIR 2) bands, including B6 and B7 in Landsat 8, where B11 and B12 in Sentinel 2 are used to evaluate moisture levels and structural differences in vegetation. By using these selected spectral bands, the high dimensional data is created, which is then split into training and testing sets. Data classification is a crucial task as it assigns different labeled categories to input data based on its features. For classifying the mangroves in order to analyze the spatial changes within the study area, three different density classes are considered such as Sparse, Moderate, and Dense mangroves. In this study, Convolutional Neural Networks (CNNs) are chosen as the classification model, leveraging their ability to automatically extract hierarchical spatial and spectral features from processed images that enable the model to identify complex patterns between spectral bands for accurate classification of mangrove and non-mangrove areas.

Proposed architecture

The core of CNN architecture is the convolutional layer. These are responsible for extracting spatial features from input image. Filters or kernels present in this layer slide over the input data. This operation captures essential spatial hierarchies such as edges, textures, and patterns, which are vital for recognizing complex structures in satellite images. Convolutional layer allows the model to learn specific features such as differences in spectral characteristics between mangrove and non-mangrove region. In our proposed architecture, we incorporated

over six convolutional layers in order to extract and refine features. Multiple convolutional layers are stacked to learn more abstract features at each level, starting from simple patterns like edges in lower layers to complex features like shapes and structures in higher layers. The model was implemented using Keras with TensorFlow backend, utilizing essential deep learning libraries such as keras, numpy, and pandas for data processing and model training. We utilized the Rectified Linear Unit (ReLU) as the activation function, mentioned in Eq. (7), which introduced non-linearity in our model by converting all the negative values to zero, enabling the network to model intricate relationships. Then we integrated the batch normalization after every convolutional layer to stabilize training process by normalizing the activations, thereby reducing internal covariate shift. This improves convergence and enables our network to train efficiently. ReLU and batch normalization together ensure robust performance in our proposed model, especially in differentiating the spectral differences between mangrove and non-mangrove regions.

$$f(x) = max(0, x) \tag{7}$$

where x is the input to the neuron. We incorporated one dropout layer after every two convolutional layers in convolutional stack to further enhance the robustness of our model. Dropout serves as a regularization technique that deactivates fraction of neurons by randomly setting them to zero during the model training that restricts our model from overly relying on specific neurons and allows the network to learn more generalized features. This arrangement is particularly effective where spatial dependencies can lead to model overfitting. By applying dropout, our proposed network maintains its ability to accurately differentiate between mangrove and non-mangrove areas.

Global max pooling layers reduce the spatial dimensions of the feature maps produced by the convolutional layers. Pooling layers perform downsampling by choosing the maximum value from each patch of feature map. To perform pooling operation across the entire feature map, we incorporated a global max pooling layer in our model. This reduces the overall size of data while retaining the important features. This is crucial for distinguishing between mangrove and non-mangrove areas at different density levels, enabling the model to focus on the most important spatial features without being affected by the spatial resolution.

Dense or fully connected layers perform the final classification. The flattened outputs from previous layers is passed through one or more dense layers combining the extracted features and learn complex relationships between them. This enables the model to label different density classes within the study area. Our model incorporates three dense layers with ReLU as their activation function, specified in Eq. (7), each followed by a dropout layer. Including dropout layer after each dense layer prevents overfitting where the model could otherwise learn overly complex patterns that don't generalize well. In the fully connected layer, softmax is applied as its activation function, given in the Eq. (8). The purpose of softmax is to convert the raw outputs from the final layer into probabilities. By normalizing the outputs, softmax ensures that the predicted values for each class sum to 1, making our model's outputs interpretable as the likelihood of each mangrove density class.

$$s(x_i) = \frac{e^{x_i}}{\sum_0^N e^{x_j}} \tag{8}$$

where, s is softmax, x is input vector, N is number of classes in the multi-class classifier, e^{x_i} and e^{x_j} are standard exponential function for input vector and output vector respectively.

Model training and parameters

In the proposed model, hyperparameters were selected to optimize the performance of our model. The first key parameter is the number of neurons used in convolutional layers are set to 128, which establishes a balance between model complexity and capacity for feature extraction. This value is increased progressively in the deeper layers with neuron values of 256 and 512 in the following convolutional layers. Dropout regularization is applied with a rate of 0.2 across the layers to prevent overfitting, particularly important in deep networks. The kernel size was set to 1 to focus on per-pixel spectral classification rather than spatial feature extraction, treating each pixel's spectral signature as an independent feature vector. This approach is useful for classifying spectral differences in remote sensing applications, where pixel-wise classification is critical. Adam optimizer is chosen for its adaptive learning rate capabilities, and categorical cross entropy is used as loss function, as specified in Eq. (9) to handle multi-class classification.

$$\mathrm{L} = -\sum_{\mathrm{i}=1}^{\mathrm{C}} y_i \log(p_i) \tag{9}$$

where L is loss, C is the number of classes, y_i is the true label for class i, which is 1 if the sample belongs to class i and 0 otherwise, p_i is the predicted probability for class i.

To evaluate the performance of the proposed CNN model, the dataset was divided into training and testing sets using an 80:20 ratio. This stratified split ensured that all three mangrove density classes sparse, moderate, and dense were proportionally represented in both sets. The random seed was fixed during the split to ensure reproducibility of the results. Once the model is done with training, the composite image is fed into the model to generate various class maps. After predicting the available classes within the study area, the class maps obtained are given in Fig. 4(a)–4(f). The model's performance is evaluated by estimating various metrics such as accuracy, F1 score, kappa coefficient. To assess the model's effectiveness and dependability in achieving

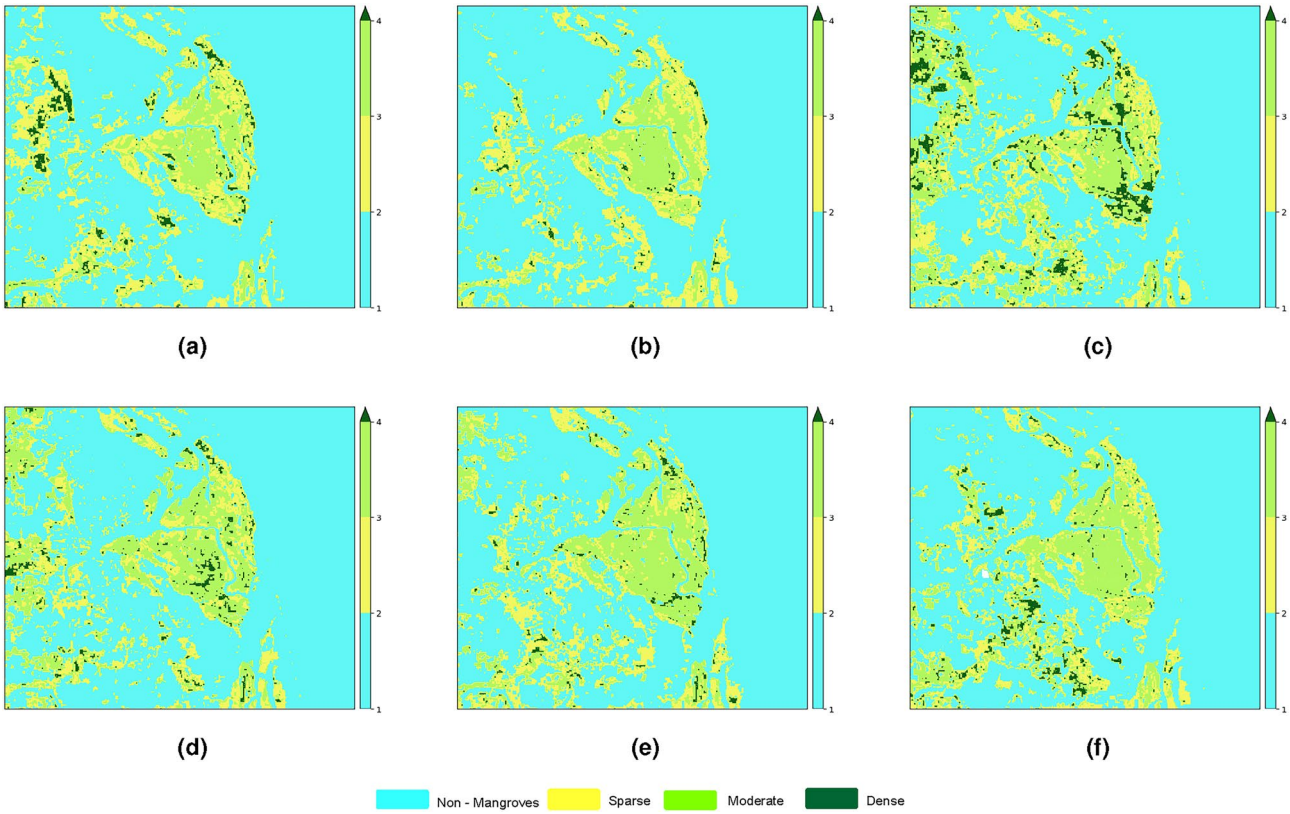


Fig. 4. Mangrove class maps (**a**) 2014, (**b**) 2016, (**c**) 2018, (**d**) 2020, (**e**) 2022, (**f**) 2024. Maps generated using Python 3.9 (https://www.python.org/) with Matplotlib 3.5.1 (https://matplotlib.org/).

accurate predictions, True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) are computed. The accuracy is calculated as given in the Eq. (10).

$$accuracy = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{10}$$

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

$$\mathrm{F1\ score} = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{13}$$

Estimating performance metrics like F1 score and kappa coefficient, along with accuracy provides more nuanced evaluation of model in terms of its performance, with F1 score balancing precision and recall for imbalanced datasets, and kappa coefficient offers insights into classification reliability. F1 score is computed as the harmonic mean of precision and recall. The precision, recall, F1 score, and kappa coefficient are calculated as given in Eqs. (11), (12), (13) and (14) respectively.

$$\mathrm{k} = \frac{P_o - P_e}{1 - P_e} \tag{14}$$

where k is kappa coefficient, P_o and P_e are observed and expected agreements.

Time series analysis

Time series analysis in this study examines the temporal changes in mangroves by computing class-specific values across sequential satellite images from 2014 to 2024. Using the trained CNN model, the proportions of sparse, moderate, and dense mangrove classes are calculated for each year. These values are organized in tabular form to observe trends in mangrove density classes and distribution over time as shown in Table 3. Tracking the extent of each class annually enables a quantitative assessment of changes in mangrove cover which analyzes effectiveness of conservation measures and identifying areas for further ecological intervention.

Year	Sparse/open canopy (%)	Moderate canopy (%)	Dense/closed canopy (%)	Total mangroves (%)	Non-mangroves (%)
2014	17.23	10.45	5.12	32.80	67.20
2015	17.55	10.75	4.52	32.82	67.18
2016	16.78	11.35	4.03	32.16	67.84
2017	16.32	12.07	6.21	34.60	65.40
2018	16.98	14.12	8.04	39.14	60.86
2019	17.11	14.52	6.53	38.16	61.84
2020	18.03	15.27	5.01	38.31	61.69
2021	17.76	14.57	5.52	37.85	62.15
2022	16.19	14.06	4.28	34.53	65.47
2023	15.50	14.55	5.04	35.09	64.91
2024	15.00	15.00	6.08	36.08	63.92

Table 3. Mangrove canopy cover by density class for each year from 2014 to 2024.

Method	Overall accuracy (%)	F1-score	Kappa coefficient
CMFE (proposed)	**94.00**	**0.91**	**0.88**
REMI[1]	89.50	0.85	0.80
GCMI[7]	88.30	0.83	0.78
Random Forest (RF)[8]	91.10	0.84	0.79
Support Vector Machine (SVM)[2]	90.20	0.82	0.76

Table 4. Performance comparison of different methods.

The sparse or open canopy show slight variations that range from 15.00% to its peak of 18.03% in 2020 before its gradual decline towards 2024. Moderate canopy coverage exhibits an upward trend from 2014 to 2024, with values increasing from 10.45% to a high of 15.00%. Dense or closed canopy cover kept constantly varying with the exception of a small increase in 2018 having a peak of 8.04%. The total mangrove canopy cover showed an increasing trend from 32.80% to 36.08% between 2014 and 2024 respectively. This trend may reflect improved environmental conditions and management interventions during certain periods. For instance, the increase observed between 2017 and 2020 aligns with the timeframe of several local conservation initiatives. Concurrently, the percentage of non-mangrove areas decreased from 67.20% in 2014 to 63.92% in 2024. This reduction may suggest a trend toward mangrove habitat stabilization or recovery.

Results

Accuracy comparison with existing index-based methods

To accurately assess the accuracy and reliability of proposed Cloud-Masked Feature Extraction (CMFE) approach, we compared it against other existing models. The CMFE approach showed an improvement of 5.7% and 4.5% as shown in Table 4, over the GCMI[7] and REMI[1], respectively. These results demonstrate CMFE's suitability for consistent and accurate mangrove density classification in the study area Pichavaram mangroves. CMFE's high F1-score and kappa coefficient underscore its robustness in effectively distinguishing sparse, moderate, and dense mangrove areas, even under challenging environmental conditions. Unlike index-based methods, which often rely on a limited set of spectral bands, the CMFE classification model integrates high-dimensional spectral and textural features, allowing for more precise classification of mangrove density. The effectiveness of the CMFE classification model is evident from its ability to capture subtle variations in mangrove density, which are often misclassified in traditional index-based approaches.

Performance metrices and comparable studies

For a detailed evaluation, we compared proposed CMFE against machine learning models, including traditional machine learning models like Random Forest (RF) and Support Vector Machine (SVM). As these models are widely used in remote sensing, they often struggle to accurately classify intricate density gradations in mangrove vegetation. The high-dimensional feature set in CMFE, covers spectral and textural information, allows for more accurate density classification than RF and SVM. This underscores proposed CMFE's effectiveness in accurately classifying mangrove density across diverse and complex real-world environments and its superior performance in classifying mangrove density. The scalability of the method was also explored through a comparative analysis using the Sundarbans mangrove forest, which differs from Pichavaram in species composition, tidal conditions, and spatial complexity. While CMFE showed strong potential, limited satellite resolution led to incomplete coverage in fragmented zones, which may cause misclassifications and reduce accuracy in health assessments across such regions. With an overall accuracy of 94.0%, CMFE outperforms RF[8] and SVM[2] by 2.9% and 3.8%, respectively. Also, CMFE achieves a higher F1-score of 0.91 and kappa coefficient of 0.88, compared to the RF's F1-score of 0.84 and kappa of 0.79, and SVM's F1-score of 0.82 and kappa of 0.76 shown in Table 4. These

improvements indicate that proposed CMFE's deep learning-based approach, which utilizes high-dimensional spectral and textural features along with effective cloud-masking, provides a more consistent and accurate classification than traditional machine learning models. The enhanced performance of proposed CMFE highlights its ability to capture finer distinctions in mangrove density especially in complex and heterogeneous mangrove ecosystems.

Conclusion

In this paper, we presented a novel methodology for analyzing spatial and temporal changes in mangrove forests in Pichavaram, India, focusing on masking out the unclear pixels and restoring the missing information along with density-based classification over a decade from 2014 to 2024. The proposed model resulted in an accuracy of 94% which highlights its robustness in classifying the mangroves into distinct density classes. These results highlight the approach's robustness in detecting and monitoring mangrove dynamics within the studied region. While this study was conducted exclusively in the Pichavaram Mangrove Forest, the promising results suggest potential for broader applicability though further validation across different mangrove habitats is needed to confirm its generalizability.

Data availability

Data is provided within the supplementary information files.

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Author contributions

Dr. Kathiroli R, as the corresponding author, provided guidance and supervision throughout the project, ensuring its successful completion. Madhumitha P S, Achyut Prasad D C, and Sandhiya S contributed equally to the implementation and analysis of this research work.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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